CT4101 Milestone 2

First Pass Data Audit

Source:

Predict Students' Dropout and Academic Success Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89

License:

This dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

Retrieval Date:

12th/October/2026

Collection Context:

The data was collected by Mónica Vieira Martins, Valentim Realinho, Luis Baptista, Jorge Machado at the Polytechnic University of Portalegre. It was collected to “contribute to the reduction of academic failure at higher education, by using machine learning techniques to identify students at risk of failure at an early stage of their academic path, so that strategies to support them can be put into place.” The data was collection from Portuguese students in a single university

No candidate identifier columns exist in this dataset, likely due to GDPR concerns.

Student Demographics

Variable Name	Role	Type	Missing Values	Description
Gender	Feature	Integer	No	1 – Male 0 – Female
Age at enrollment	Feature	Integer	No	Age of Student at enrollment
Marital status	Feature	Integer	No	1 – single 2 – married 3 – widower 4 – divorced 5 – facto union 6 – legally separated
Nationality	Feature	Integer	No	1 - Portuguese 2 - German 6 - Spanish 11 - Italian 13 - Dutch 14 - English 17 - Lithuanian 21 - Angolan 22 - Cape Verdean 24 - Guinean 25 - Mozambican 26 - Santomean 32 - Turkish 41 - Brazilian 62 - Romanian 100 - Moldavian 101 - Mexican 103 - Ukrainian 105 - Russian 108 - Cuban 109 - Colombian
Displaced	Feature	Integer	No	1 – yes 0 – no
Educational special needs	Feature	Integer	No	1 – yes 0 – no
International	Feature	Integer	No	1 – yes 0 – no

Student Parental Information

Variable Name	Role	Type	Missing Values	Description
Father's occupation	Feature	Integer	No	0 - Student 1 - Representatives of the Legislative Power and Executive Bodies, Directors, Directors and Executive Managers 2 - Specialists in Intellectual and Scientific Activities 3 - Intermediate Level Technicians and Professions 4 - Administrative staff 5 - Personal Services, Security and Safety Workers and Sellers 6 - Farmers and Skilled Workers in Agriculture, Fisheries and Forestry 7 - Skilled Workers in Industry, Construction and Craftsmen

				8 - Installation and Machine Operators and Assembly Workers 9 - Unskilled Workers 10 - Armed Forces Professions 90 - Other Situation 99 - (blank) 101 - Armed Forces Officers 102 - Armed Forces Sergeants 103 - Other Armed Forces personnel 112 - Directors of administrative and commercial services 114 - Hotel, catering, trade and other services directors 121 - Specialists in the physical sciences, mathematics, engineering and related techniques 122 - Health professionals 123 - teachers 124 - Specialists in finance, accounting, administrative organization, public and commercial relations 131 - Intermediate level science and engineering technicians and professions 132 - Technicians and professionals, of intermediate level of health 134 - Intermediate level technicians from legal, social, sports, cultural and similar services 135 - Information and communication technology technicians 141 - Office workers, secretaries in general and data processing operators 143 - Data, accounting, statistical, financial services and registry-related operators 144 - Other administrative support staff 151 - personal service workers 152 - sellers 153 - Personal care workers and the like 154 - Protection and security services personnel 161 - Market-oriented farmers and skilled agricultural and animal production workers 163 - Farmers, livestock keepers, fishermen, hunters and gatherers, subsistence 171 - Skilled construction workers and the like, except electricians 172 - Skilled workers in metallurgy, metalworking and similar 174 - Skilled workers in electricity and electronics 175 - Workers in food processing, woodworking, clothing and other industries and crafts 181 - Fixed plant and machine operators 182 - assembly workers 183 - Vehicle drivers and mobile equipment operators 192 - Unskilled workers in agriculture, animal production, fisheries and forestry 193 - Unskilled workers in extractive industry, construction, manufacturing and transport 194 - Meal preparation assistants 195 - Street vendors (except food) and street service providers
Mother's occupation	Feature	Integer	No	0 - Student 1 - Representatives of the Legislative Power and Executive Bodies, Directors, Directors and Executive Managers 2 - Specialists in Intellectual and Scientific Activities 3 - Intermediate Level Technicians and Professions 4 - Administrative staff

				5 - Personal Services, Security and Safety Workers and Sellers 6 - Farmers and Skilled Workers in Agriculture, Fisheries and Forestry 7 - Skilled Workers in Industry, Construction and Craftsmen 8 - Installation and Machine Operators and Assembly Workers 9 - Unskilled Workers 10 - Armed Forces Professions 90 - Other Situation 99 - (blank) 122 - Health professionals 123 - teachers 125 - Specialists in information and communication technologies (ICT) 131 - Intermediate level science and engineering technicians and professions 132 - Technicians and professionals, of intermediate level of health 134 - Intermediate level technicians from legal, social, sports, cultural and similar services 141 - Office workers, secretaries in general and data processing operators 143 - Data, accounting, statistical, financial services and registry-related operators 144 - Other administrative support staff 151 - personal service workers 152 - sellers 153 - Personal care workers and the like 171 - Skilled construction workers and the like, except electricians 173 - Skilled workers in printing, precision instrument manufacturing, jewelers, artisans and the like 175 - Workers in food processing, woodworking, clothing and other industries and crafts 191 - cleaning workers 192 - Unskilled workers in agriculture, animal production, fisheries and forestry 193 - Unskilled workers in extractive industry, construction, manufacturing and transport 194 - Meal preparation assistants
Father's qualification	Feature	Integer	No	1 - 12th Year of Schooling or Equiv. 2 - Bachelor's Degree 3 - Degree 4 - Master's 5 - Doctorate 6 – Frequency of Higher Education 9 - 12th Year of Schooling (Not Completed) 10 - 11th Year of Schooling (Not Completed) 11 - 7th Year (Old) 12 – Other, 11th Year of Schooling 13 - 2nd year complementary high school course 14 - 10th Year of Schooling 18 - General commerce course 19 - Basic Education 3rd Cycle (9th/10th/11th Year) or Equiv. 20 - Complementary High School 22 - Technical-professional course 25 - Complementary High School Course - not concluded 26 - 7th year of schooling 27 - 2nd cycle of the general high school course 29 - 9th Year of Schooling - Not Completed 30 - 8th year of schooling 31 - General Course of Administration and Commerce

				33 - Supplementary Accounting and Administration 34 - Unknown 35 - Can't read or write 36 - Can read without having a 4th year of schooling 37 - Basic education 1st cycle (4th/5th year) or equiv. 38 - Basic Education 2nd Cycle (6th/7th/8th Year) or Equiv. 39 - Technological specialization course 40 - Higher education - degree (1st cycle) 41 - Specialized higher studies course 42 - Professional higher technical course 43 - Higher Education - Master (2nd cycle) 44 - Higher Education - Doctorate (3rd cycle)
Mother's qualification	Feature	Integer	No	1 - 12th Year of Schooling or Equiv. 2 - Bachelor's Degree 3 - Degree 4 - Master's 5 - Doctorate 6 – Frequency of Higher Education 9 - 12th Year of Schooling (Not Completed) 10 - 11th Year of Schooling (Not Completed) 11 - 7th Year (Old) 12 – Other, 11th Year of Schooling 14 - 10th Year of Schooling 18 - General commerce course 19 - Basic Education 3rd Cycle (9th/10th/11th Year) or Equiv. 22 - Technical-professional course 26 - 7th year of schooling 27 - 2nd cycle of the general high school course 29 - 9th Year of Schooling - Not Completed 30 - 8th year of schooling 34 - Unknown 35 - Can't read or write 36 - Can read without having a 4th year of schooling 37 - Basic education 1st cycle (4th/5th year) or equiv. 38 - Basic Education 2nd Cycle (6th/7th/8th Year) or Equiv. 39 - Technological specialization course 40 - Higher education - degree (1st cycle) 41 - Specialized higher studies course 42 - Professional higher technical course 43 - Higher Education - Master (2nd cycle) 44 - Higher Education - Doctorate (3rd cycle)

Student Finances

Variable Name	Role	Type	Missing Values	Description
Scholarship holder	Feature	Integer	No	1 – yes 0 – no
Tuition fees up to date	Feature	Integer	No	1 – yes 0 – no
Debtor	Feature	Integer	No	1 – yes 0 – no

Student Application

Variable Name	Role	Type	Missing Values	Description
Course	Feature	Integer	No	33 - Biofuel Production Technologies 171 - Animation and Multimedia Design 8014 - Social Service (evening attendance) 9003 - Agronomy 9070 - Communication Design 9085 - Veterinary Nursing 9119 - Informatics Engineering 9130 - Equinculture 9147 - Management 9238 - Social Service 9254 - Tourism 9500 - Nursing 9556 - Oral Hygiene 9670 - Advertising and Marketing Management 9773 - Journalism and Communication 9853 - Basic Education 9991 - Management (evening attendance)
Admission grade	Feature	Continuous	No	Admission grade (between 0 and 200)
Application mode	Feature	Integer	No	1 - 1st phase - general contingent 2 - Ordinance No. 612/93 5 - 1st phase - special contingent (Azores Island) 7 - Holders of other higher courses 10 - Ordinance No. 854-B/99 15 - International student (bachelor) 16 - 1st phase - special contingent (Madeira Island) 17 - 2nd phase - general contingent 18 - 3rd phase - general contingent 26 - Ordinance No. 533-A/99, item b2) (Different Plan) 27 - Ordinance No. 533-A/99, item b3 (Other Institution) 39 - Over 23 years old 42 - Transfer 43 - Change of course 44 - Technological specialization diploma holders 51 - Change of institution/course

				53 - Short cycle diploma holders 57 - Change of institution/course (International)
Application order	Feature	Integer	No	Application order (between 0 - first choice; and 9 last choice)
Daytime/evening attendance	Feature	Integer	No	1 – daytime 0 - evening

Student Academics

Variable Name	Role	Type	Missing Values	Description
Previous qualification	Feature	Integer	No	1 - Secondary education 2 - Bachelor's 3 - Degree 4 – Master's 5 - Doctorate 6 - Frequency of higher education 9 - 12th year of schooling (not completed) 10 - 11th year of schooling (not completed) 12 – Other, 11th year of schooling 14 - 10th year of schooling 15 - 10th year of schooling (not completed) 19 - Basic education 3rd cycle or equiv. 38 - Basic education 2nd cycle or equiv. 39 - Technological specialization course 40 - Higher education degree (1st cycle) 42 - Professional higher technical course 43 - Master (2nd cycle)
Previous qualification (grade)	Feature	Continuous	No	Grade of previous qualification (between 0 and 200)
Curricular units 1st sem (credited)	Feature	Integer	No	Number of curricular units credited in the 1st semester
Curricular units 1st sem (enrolled)	Feature	Integer	No	Number of curricular units enrolled in the 1st semester

Curricular units 1st sem (evaluations)	Feature	Integer	No	Number of evaluations to curricular units in the 1st semester
Curricular units 1st sem (approved)	Feature	Integer	No	Number of curricular units approved in the 1st semester
Curricular units 1st sem (grade)	Feature	Integer	No	Grade average in the 1st semester (between 0 and 20)
Curricular units 1st sem (without evaluations)	Feature	Integer	No	Number of curricular units without evalutions in the 1st semester
Curricular units 2nd sem (credited)	Feature	Integer	No	Number of curricular units credited in the 2nd semester
Curricular units 2nd sem (enrolled)	Feature	Integer	No	Number of curricular units enrolled in the 2nd semester
Curricular units 2nd sem (evaluations)	Feature	Integer	No	Number of evaluations to curricular units in the 2nd semester
Curricular units 2nd sem (approved)	Feature	Integer	No	Number of curricular units approved in the 2nd semester
Curricular units 2nd sem (grade)	Feature	Integer	No	Grade average in the 2nd semester (between 0 and 20)
Curricular units 2nd sem (without evaluations)	Feature	Integer	No	Number of curricular units without evalutions in the 1st semester

National Information

Variable Name	Role	Type	Missing Values	Description
Unemployment rate	Feature	Continuous	No	Unemployment rate (%)
Inflation rate	Feature	Continuous	No	Inflation rate (%)
GDP	Feature	Continuous	No	GDP

Target

Variable Name	Role	Type	Missing Values	Description
Target	Target	Categorical	No	The problem is formulated as a three category classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course

Target Distribution

Category	Quantity	Percentage
Graduate	2209	49.93%
Dropout	1421	32.12%
Enrolled	794	17.95%
Total	4424	100%

Training/Test Split

The testing and training data is split using Train and Test Split.py and it splits 80% of the data for training, and 20% for testing, and it ensures that the target distribution of both data sets is roughly equivalent, both with eachother and with the original dataset.

The split of 80/20 was chosen so the majority of the data can be used to test the model, while leaving 20% left (~900 records) to test the model and see what improvements have been made over the baseline model.

Baseline Model Results

The baseline, which was just predicting the most common target value (e.g. Graduated) had an accuracy of 49.94%, and a precision of 49.94% for Graduate, with 0% for Dropout and Enrolled.

Repository Link

https://github.com/EoinCorless/CT4101_Student_dropout_and_success